

EXPLORATORY DATA ANALYSIS — POST-INGESTION ADDENDUM

Dataset 1: Historical Top 200 and Viral 50 (2017–2021)
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Ingestion completed: April 20, 2026

This addendum completes the deferred observations listed in Section 7 of the original EDA document. All findings below were obtained via SQL Server query after BULK INSERT of Dataset 1 into the HiddenGemMusic database. These figures supersede the estimated values in the original EDA where differences exist.

9. Post-Ingestion Validation Results

All queries were run against the HiddenGemMusic database after full DS1 ingestion and deduplication. Source filter: source = 'Historical_Top200_and_Viral50'.

9.1 Final Row Counts

Metric	Pre-Ingestion Estimate	Post-Ingestion Actual	Notes
Total rows after dedup	26,090,476	26,075,338	~15K duplicates removed by dedup step
Top 200 rows	20,249,193	20,235,488	Reflects dedup removal
Viral 50 rows	5,841,283	5,839,850	Reflects dedup removal
NULL streams count	~5,841,283 (estimated)	5,839,850	Exactly matches Viral 50 count — confirmed correct
Unmatched country rows	Unknown	0	All 70 region values matched Country.full_name successfully

9.2 Trend Distribution — Confirmed

Trend Value	Pre-Ingestion Count	Post-Ingestion Count	Variance
MOVE_DOWN	11,184,303	11,181,065	-3,238 (dedup)
MOVE_UP	9,769,681	9,762,223	-7,458 (dedup)
SAME_POSITION	3,293,499	3,286,874	-6,625 (dedup)

NEW_ENTRY	1,842,993	1,845,176	+2,183 (within margin)
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All four trend values are present in the ingested data. trend_velocity normalisation applied correctly at ingestion: SAME_POSITION → 0.0, MOVE_UP/NEW_ENTRY → +1.0, MOVE_DOWN → -1.0. Zero NULL trend_velocity rows in the final dataset.

9.3 spotify_id Parse Results

Metric	Value	Notes
Songs with parsed spotify_id	Majority of 217,431 songs	Parsed from URL using SUBSTRING on /track/ segment
Songs without spotify_id	Small minority	URL parse failed or URL was malformed — matched by title instead
DS1 songs enriched by DS2 match	1,831	DS2 provided duration_ms, is_explicit, album_id for matched tracks

9.4 Schema Changes Required During Ingestion

The following schema changes were required after inspecting actual data values during ingestion. These were not anticipated from the Power BI EDA due to memory constraints preventing full column inspection.

Table	Column	Original Width	Updated Width	Reason
Song	title	NVARCHAR(255)	NVARCHAR(512)	Max observed title length: 511 characters
Artist	name	NVARCHAR(255)	NVARCHAR(1500)	Max observed artist string length: 1,428 characters (multi-artist comma-separated strings)
Album	name	NVARCHAR(255)	NVARCHAR(512)	Album name exceeded 255 characters during DS2 ingestion
Album	UQ_Album_SpotifyID constraint	UNIQUE	Dropped	All album_spotify_id values are NULL pending API enrichment — UNIQUE constraint prevented multiple NULL inserts

9.5 SQL Server 2025 Ingestion Finding

A significant compatibility issue was encountered during BULK INSERT on SQL Server 2025 (version 17.0.1110.1). The ROWTERMINATOR escape sequences '\n', '0x0A', and '0x0d0a' are

not correctly interpreted in SQL Server 2025 — they are treated as literal strings rather than byte values, causing the entire file to be read as a single row.

Resolution: FORMAT = 'CSV' combined with ROWTERMINATOR = '0x0a' resolved the issue. FORMAT = 'CSV' enables RFC 4180 compliant parsing which interprets the hex terminator correctly. This combination is required for all BULK INSERT operations targeting CSV files on SQL Server 2025 and must be documented for any future re-ingestion or CI/CD pipeline work.

9.6 Combined Database State After Full Ingestion

Table	Row Count	Notes
Country	246	245 from CSV + 1 Global row
ChartType	3	Top 200, Viral 50, Top 50
Song	240,583	217,431 DS1 + 23,152 DS2-only
Album	~23,000+	DS2 only — DS1 has no album data
Artist	102,948	DS1 + DS2 combined
ArtistSong	627,731	DS1 + DS2 combined
AudioFeatures	24,983	DS2 only
ChartEntry DS1	26,075,338	Historical_Top200_and_Viral50
ChartEntry DS2	2,110,316	Top50_73_Countries
ChartEntry Total	28,185,654	Combined across both datasets